本文引用:缑梦帆,和俊杰.针刺联合计算机言语疾病矫治训练治疗儿童孤独症疗效观察[J].新乡医学院学报,2024,41(1):26-31.DOI:10.7683/xxyxyxb.2024.01.005.

【临床研究】

针刺联合计算机言语疾病矫治训练治疗儿童孤独症疗效观察

缑梦帆,和俊杰
(新乡医学院附属濮阳市油田总医院儿科,河南　濮阳　457000)

摘要:　目的　探讨针刺联合计算机言语疾病矫治训练治疗孤独症患儿的临床疗效。**方法**　选择2019年12月至2021年12月新乡医学院附属濮阳市油田总医院儿科收治的108例孤独症患儿为研究对象,根据治疗方式将患儿分为对照组和观察组,每组54例。对照组患儿接受计算机言语疾病矫治训练,观察组患儿在对照组治疗基础上联合针刺治疗,所有患儿治疗4个月。分别于治疗前后采用儿童孤独症评定量表(CARS)评估患儿的症状严重程度,采用语言发育迟缓检查法(S-S法)评估患儿的语言功能治疗效果,采用孤独症儿童行为量表(ABC)评估患儿的异常行为,采用格塞尔发育量表(GDS)评估患儿的中枢神经系统功能。采用重危患者家属满意度量表中文修订版(C-CCFSS)评估患儿家属满意度。**结果**　治疗前2组患儿的CARS评分比较差异无统计学意义($P>0.05$);治疗后,观察组患儿的CARS评分显著低于对照组($P<0.05$);对照组患儿治疗前后的CARS评分比较差异无统计学意义($P>0.05$);观察组患儿治疗后的CARS评分显著低于治疗前($P<0.05$)。对照组和观察组患儿的语言功能治疗总有效率分别为83.33%(45/54)、96.30%(52/54),观察组患儿的语言功能治疗总有效率显著高于对照组($\chi^2=4.960$,$P<0.05$)。治疗前2组患儿的ABC各项评分及总分比较差异无统计学意义($P>0.05$);2组患儿治疗后的ABC各项评分及总分显著低于治疗前($P<0.05$);治疗后,观察组患儿的ABC各项评分及总分显著低于对照组($P<0.05$)。治疗前2组患儿的GDS评分比较差异无统计学意义($P>0.05$);治疗后,观察组患儿的GDS评分显著高于对照组($P<0.05$);对照组患儿治疗前后的GDS评分比较差异无统计学意义($P>0.05$);观察组患儿治疗后的GDS评分显著高于治疗前($P<0.05$)。对照组和观察组患儿家属的满意度分别为81.48%(44/54)、94.44%(51/54),观察组患儿家属的满意度显著高于对照组($\chi^2=4.285$,$P<0.05$)。**结论**　针刺联合计算机言语疾病矫治训练能够有效改善孤独症患儿临床症状及异常行为严重程度,促进语言功能及智力恢复,家属满意度较高,值得临床推广应用

关键词:　孤独症;针刺;计算机言语疾病矫治训练;言语康复;疗效
中图分类号:R729　**文献标志码:**A　**文章编号:**1004-7239(2024)01-0026-06

Clinical efficacy of acupuncture combined with computer speech disorder correction training in the treatment of autistic children

GOU Mengfan, HE Junjie
(*Department of Pediatrics, Puyang Oilfield General Hospital Affiliated to Xinxiang Medical University, Puyang* 457000, *Henan Province, China*)

Abstract:　Objective　To explore the clinical efficacy of acupuncture combined with computer speech disorder correction training in the treatment of children with autism. **Methods**　A total of 108 children with autism admitted to the Department of Pediatrics, Puyang Oilfield General Hospital Affiliated to Xinxiang Medical University from December 2019 to December 2021 were selected as the research subjects. According to the treatment methods, the children were divided into the control group and the observation group, with 54 children in each group. The children in the control group received computer speech disorder correction training, while the children in the observation group received acupuncture treatment on the basis of the control group, and all children were treated for 4 months. Before and after treatment, the severity of symptoms of children was evaluated by the childhood autism rating scale (CARS), the treatment effectiveness of language function of children was evaluated by the language delayed development test (S-S method), the abnormal behavior of children was evaluated by the autism behavior checklist (ABC), and the central nervous system function of children was evaluated by the Gesell development scale (GDS). The satisfaction of the children's family members was evaluated by the Chinese version of critical care family satisfaction survey (C-CCFSS). **Results**　There was no significant difference in the CARS score of children between the two groups before treatment ($P>0.05$); after treatment, the CARS score of children in the observation group was significantly lower than that in

DOI:10.7683/xxyxyxb.2024.01.005
收稿日期:2023-02-20
作者简介:缑梦帆(1983-),女,河南濮阳人,学士,副主任医师,研究方向:儿童神经康复。

(C)1994-2024 China Academic Journal Electronic Publishing House. All rights reserved.　http://www.cnki.net

the control group ($P<0.05$); there was no significant difference in the CARS score before and after treatment in the control group ($P>0.05$); the CARS score of children in the observation group after treatment was significantly lower than that before treatment ($P<0.05$). The total effective rates of language function therapy of children in the control group and observation group were 83.33% (45/54) and 96.30% (52/54), respectively; the total effective rate of language function therapy of children in the observation group was significantly higher than that in the control group ($\chi^2=4.960$, $P<0.05$). Before treatment, there was no significant difference in the each item scores and scores total scores of ABC of children between the two groups ($P>0.05$); the each item scores and total scores of ABC of children in the two groups after treatment were significantly lower than those before treatment ($P<0.05$); after treatment, the each item scores and total scores of ABC of children in the observation group were significantly lower than those in the control group ($P<0.05$). Before treatment, there was no significant difference in GDS score of children between the two groups ($P>0.05$); after treatment, the GDS score of children in the observation group was significantly higher than that in the control group ($P<0.05$); there was no significant difference in GDS score of children before and after treatment in the control group ($P>0.05$); the GDS score of children in the observation group after treatment was significantly higher than that before treatment ($P<0.05$). The satisfaction of family member of children in the control group and the observation group was 81.48% (44/54) and 94.44% (51/54), respectively. The satisfaction of family member of children in the observation group was significantly higher than that in the control group ($\chi^2=4.285$, $P<0.05$). **Conclusion** Acupuncture combined with computer speech disorder correction training can effectively improve the severity of the clinical symptoms and abnormal behavior of children with autism, promote language function and intellectual recovery, and has high family satisfaction, which is worthy of clinical promotion and application.

Key words: autism; acupuncture; computer speech disorder correction training; speech rehabilitation; curative effect

孤独症又称孤独性障碍或自闭症,属于一种发育障碍疾病[1]。患儿多于发育早期出现各种孤独症症状,具体表现为交流障碍、行为异常、兴趣狭窄等,部分患儿伴随不同程度智力障碍[2]。由于患儿存在社交、言语、智力障碍,其日常生活能力往往较差。孤独症儿童的言语障碍不仅影响儿童认知能力,还对其日常生活和社交能力产生巨大影响,日常生活中易出现各种问题[3]。另外,孤独症患儿预后较差,多数患儿缺乏自理能力,成年后仍需照顾,难以独立生活,给家庭及社会带来沉重负担。目前,对于孤独症缺乏有效的临床治疗手段,常规治疗手段为通过康复治疗和教育改善患儿各种障碍,但收效甚微。随着现代医学技术及科技技术的发展进步,计算机系统被应用于各种疾病的临床治疗。有研究发现,计算机言语疾病矫治训练通过运用多媒体技术能有效改善失语症患者的言语功能[4]。近年来针刺因具有操作简便、经济安全的特点而广泛应用于临床多种疾病的治疗[5-6]。有研究发现,针刺通过刺激头部或四肢相应穴位,改善局部血流灌注,从而改善孤独症患儿的情绪、睡眠障碍[7]。但针刺联合计算机言语疾病矫治训练在治疗孤独症中的临床疗效尚无定论。基于此,本研究旨在探讨针刺联合计算机言语疾病矫治训练治疗孤独症患儿的临床疗效,以期为孤独症患儿的治疗提供参考。

1　资料与方法

1.1　一般资料

选择 2019 年 12 月至 2021 年 12 月新乡医学院附属濮阳市油田总医院儿科收治的孤独症患儿为研究对象。病例纳入标准:(1)符合《孤独症谱系障碍儿童早期识别筛查和早期干预专家共识》[8]中孤独症的相关诊断标准;(2)3 岁前在多种场合下出现刻板重复的行为方式、兴趣范围狭窄、社会交往障碍、言语发育障碍等典型症状者;(3)根据患儿全面的生长发育史、病史及精神检查确诊为孤独症者;(4)视觉、听觉正常者;(5)患儿家属或监护人对本研究知情同意并签署知情同意书。排除标准:(1)确诊为儿童精神分裂症、脆性 X 染色体综合征、21-三体综合征及其他神经精神障碍(如癫痫)者;(2)入院前使用过抗精神类药物治疗者;(3)治疗穴位局部有皮肤溃烂、手术瘢痕等;(4)合并重度以上智力障碍者。本研究共纳入孤独症患儿 108 例,根据治疗方式将患儿分为对照组和观察组,每组 54 例。对照组:男 38 例,女 16 例;年龄 2 ~ 9(5.13 ± 1.23)岁。观察组:男 37 例,女 17 例;年龄 2 ~ 8(5.33 ± 1.21)岁。2 组患儿的性别、年龄比较差异无统计学意义($P>0.05$),具有可比性。本研究经医院医学伦理委员会审核通过。

1.2　治疗方法

对照组患儿接受计算机言语疾病矫治训练,具体如下:在康复科医师指导下,在安静环境中使用言语障碍诊治仪(江苏常州钱璟康复器材有限公司)对孤独症患儿进行治疗,包括预备训练、指定发音训练及自选发音训练。(1)预备训练:包括基本发音训练及词、图阅读理解训练,患儿根据电脑屏幕上光点行程轨迹进行发音训练,词、图阅读理解训练中屏

(C)1994-2024 China Academic Journal Electronic Publishing House. All rights reserved.　http://www.cnki.net

幕显示词词匹配及词图匹配,计算机给出匹配结果正确与否的视觉反馈信息。(2)指定发音训练:当用键盘选中发音内容后,由计算机给出视觉和听觉提示,前者为发音器官示意图(声道侧视图、口形图、舌位图),后者为对应的合成语音输出,患儿根据上述提示练习发音,计算机根据患儿发音进行反馈,反馈内容为患儿发音的正确程度评估。(3)自选发音训练:首先由计算机显示数个发音内容的视觉提示,其次由患儿根据以上视觉提示自由选择其中之一发音,计算机进行实时语音识别后给出类似于指定发音训练的听觉、视觉反馈。每周治疗 5 d,每日 1 次,每次 30 min,持续治疗 4 个月。

观察组患儿在对照组基础上增加针刺治疗,具体如下:使用一次性无菌针灸针(0.35 mm×25.00 mm,苏州针灸用品有限公司),选取舌三针、本神、神庭、通里、百会、言语一区、言语二区、言语三区,头针平刺进针,沿皮刺入帽状腱膜下 3.0 ~4.5 cm,选取百会、神庭、本神、言语一区、言语二区、言语三区,平刺进针 1.5 ~2.5 cm,留针 1 h,20 min 捻转行针 1 次,每日 1 次,每周治疗 5 d,持续治疗 4 个月。

1.3 观察指标

(1)症状严重程度:分别于治疗前、后,采用儿童孤独症评定量表(childhood autism rating scale, CARS)[9]评估患儿症状的严重程度,该量表共包含 15 个项目,总分 60 分,CARS 评分 <30 分为非孤独症,30 分≤CARS 评分 <36 分为轻至中度孤独症,CARS 评分≥36 分为重度孤独症。(2)语言功能治疗效果:采用语言发育迟缓检查法(S-S 法)[10]评估患儿治疗前后的语言功能,根据评估结果判断治疗效果。显效:语言功能在原来基础上提高 2 个阶段;有效:语言功能提高 1 个阶段;无效:语言功能未提高。总有效率 =(显效例数 + 有效例数)/总例数 × 100%。(3)异常行为症状严重程度:分别于治疗前、后,采用孤独症儿童行为量表(autism behavior checklist, ABC)[11]对患儿异常行为进行评估,在评估师指导下,由与患儿共同生活 3 个月以上家属完成该量表,内容包括自我照顾、交往、语言、运动、感觉 5 个方面,共 57 个条目,得分越低说明患儿异常行为症状越轻。(4)智力发育状况:采用格塞尔发育量表(Gesell development schedule, GDS)[12]评估患儿治疗前后的智力发育状况,≥86 分为正常,76 ~ 85 分为边缘状态,≤75 分为发育迟缓。(5)家属满意度:根据重危患者家属满意度量表中文修订版(Chinese version of critical care family satisfaction survey, C-CCFSS)[13]评估患儿家属满意度,得分越高表示患儿家属对于患儿治疗阶段的满意程度越高。根据分数将患儿家属满意度分为非常满意(> 140 分),满意(105 ~ ≤140 分),一般(75 ~ ≤105 分)以及不满意(≤75 分)。满意度 =(非常满意例数 + 满意例数 + 一般满意例数)/总例数 × 100%。

1.4 统计学处理

应用 SPSS 25.0 软件进行数据统计与分析。符合正态分布的计量资料以均数 ± 标准差($\bar{x} \pm s$)表示,2 组间比较采用 t 检验;计数资料以例数和百分率表示,2 组间比较采用 χ^2 检验;$P < 0.05$ 为差异有统计学意义。

2 结果

2.1 2 组患儿症状严重程度比较

治疗前 2 组患儿的 CARS 评分比较差异无统计学意义($P > 0.05$)。治疗后,观察组患儿的 CARS 评分显著低于对照组,差异有统计学意义($P < 0.05$)。对照组患儿治疗前后的 CARS 评分比较差异无统计学意义($P > 0.05$)。观察组患儿治疗后的 CARS 评分显著低于治疗前,差异有统计学意义($P < 0.05$)。结果见表 1。

表 1 2 组患儿 CARS 评分比较

Tab. 1 Comparison of CARS score of children between the two groups ($\bar{x} \pm s$)

组别	n	CARS 评分		t	P
		治疗前	治疗后		
对照组	54	36.94 ± 3.88	35.78 ± 3.66	1.559	>0.05
观察组	54	37.09 ± 3.49	33.13 ± 3.52	5.968	<0.05
t		0.209	3.831		
P		>0.05	<0.05		

2.2 2 组患儿语言功能治疗效果比较

对照组患儿中语言功能治疗效果显效 24 例,有效 21 例,无效 9 例,总有效率为 83.33%(45/54)。观察组患儿中语言功能治疗效果显效 34 例,有效 18 例,无效 2 例,总有效率为 96.30%(52/54)。观察组患儿的语言功能治疗总有效率显著高于对照组,差异有统计学意义($\chi^2 = 4.960$, $P < 0.05$)。

2.3 2 组患儿异常行为症状严重程度比较

治疗前 2 组患儿的 ABC 各项评分及总分比较差异无统计学意义($P > 0.05$)。2 组患儿治疗后的 ABC 各项评分及总分显著低于治疗前,差异有统计学意义($P < 0.05$)。治疗后,观察组患儿的 ABC 各项评分及总分显著低于对照组,差异有统计学意义($P < 0.05$)。结果见表 2。

(C)1994-2024 China Academic Journal Electronic Publishing House. All rights reserved. http://www.cnki.net

表 2　2 组患儿 ABC 评分比较

Tab. 2　Comparison of ABC score of children between the two groups　($\bar{x} \pm s$)

组别	n	ABC 评分					
		自我照顾	交往	语言	运动	感觉	总分
对照组	54						
治疗前		24.39 ± 1.47	30.15 ± 4.15	26.15 ± 2.20	19.25 ± 2.37	35.25 ± 6.24	135.19 ± 10.27
治疗后		12.65 ± 1.23[a]	25.14 ± 2.80[a]	22.30 ± 1.47[a]	15.25 ± 1.60[a]	27.52 ± 4.15[a]	102.86 ± 5.91[a]
观察组	54						
治疗前		24.63 ± 1.91	30.27 ± 4.20	26.07 ± 2.16	19.53 ± 2.41	35.02 ± 5.92	135.52 ± 10.38
治疗后		10.24 ± 1.15[ab]	20.30 ± 2.13[ab]	18.62 ± 1.13[ab]	10.20 ± 1.25[ab]	21.07 ± 3.16[ab]	80.43 ± 4.65[ab]

注:与治疗前比较[a] $P < 0.05$;与对照组比较[b] $P < 0.05$。

2.4　2 组患儿智力发育状况比较

治疗前 2 组患儿的 GDS 评分比较差异无统计学意义($P > 0.05$)。治疗后,观察组患儿的 GDS 评分显著高于对照组,差异有统计学意义($P < 0.05$)。对照组患儿治疗前后的 GDS 评分比较差异无统计学意义($P > 0.05$);观察组患儿治疗后的 GDS 评分显著高于治疗前,差异有统计学意义($P < 0.05$)。结果见表 3。

表 3　2 组患儿 GDS 评分比较

Tab. 3　Comparison of GDS score of children between the two groups　($\bar{x} \pm s$)

组别	n	GDS 评分		t	P
		治疗前	治疗后		
对照组	54	49.35 ± 4.39	51.02 ± 5.20	1.952	> 0.05
观察组	54	49.57 ± 4.55	55.06 ± 5.62	6.264	< 0.05
t		0.258	3.872		
P		> 0.05	< 0.05		

2.5　2 组患儿家属的满意度比较

对照组患儿家属中非常满意 24 例,满意 11 例,一般满意 9 例,不满意 10 例,满意度为 81.48%(44/54)。观察组患儿家属中非常满意 38 例,满意 10 例,一般满意 3 例,不满意 3 例,满意度为 94.44%(51/54)。观察组患儿家属的满意度显著高于对照组,差异有统计学意义($\chi^2 = 4.285$, $P < 0.05$)。

3　讨论

近年来,我国孤独症发病率呈上升趋势,孤独症患儿得到了社会各界的广泛关注[14]。大多数患儿在确诊孤独症的同时发现患有言语障碍、智力障碍等问题[15]。目前,临床上常规治疗孤独症的方案多为药物治疗、锻炼患儿发音和行为疗法,但此类治疗手段多为被动手段,对孤独症患儿疗效一般。因此,探索有效治疗手段是当前临床研究的热点。

以往对孤独症患儿的治疗多基于对患儿进行长时间、反复的言语训练,帮助提高患儿理解及言语表达能力[16]。此种治疗方案具有一定疗效,但对于医患双方来说,治疗过程非常枯燥和单调,且单一依靠康复治疗师已无法满足临床需求。计算机言语疾病矫治训练以计算机技术为基础,通过使用多媒体设备,将语言康复训练程序化,使患者在定期、连续的治疗中获益[17]。同时,计算机可智能模糊识别患儿语言参数,提供个体化康复处方,孤独症患儿通过反复学习和练习,刺激控制语言中枢的神经元突触,进行语言功能区域网络重塑,调节神经元活动,促进言语功能恢复,与常规康复训练相比,针对性更强。但该方法对于依从性较差患儿的治疗效果不甚理想,因此,往往需要联合其他治疗方案。中医治疗疾病基于整体、辨证论治的理念,在各种疾病的治疗中有显著优势。孤独症在中医学中无相关病名,但孤独症患儿症状与中医中“语迟”“无慧”“目无情”等相似,即患儿的脑部发育存在一定智力障碍,进而影响其语言交流功能,多于幼儿期发病[18]。目前,中医多认为孤独症病机病因多为先天胎禀不足,后天失于调养,以致精血不足,脑髓失充,精明之府失养,即为心神迷惑之症,其病位在脑,与心、脾、肝、肾有密切关系[19]。因此,在脏腑经络理论指导下,中医可通过针刺头部对应穴位,发挥通经活络、健脑益髓的功效,进而改善孤独症患儿的神志状态及言语功能。

本研究结果显示,观察组患儿治疗后的 CARS 评分显著低于治疗前,ABC 各项评分及总分显著低于治疗前,GDS 评分显著高于治疗前;这说明,针刺联合计算机言语疾病矫治训练可显著改善孤独症患儿的症状严重程度、异常行为及智力发育状况。此外,治疗后,观察组患儿的 CARS 评分显著低于对照组,语言功能治疗总有效率显著高于对照组;ABC 各项评分及总分显著低于对照组,GDS 评分显著高于对照组,患儿家属的满意度显著高于对照组;这说明,针刺联合计算机言语疾病矫治训练在改善患儿症状严重程度、语言功能、异常行为、智力发育状况及患儿家属满意度方面的效果优于单一进行计算机言语疾病矫治训练。分析其原因为:头为诸阳之会,针刺头部穴位既可调整五脏六腑之经气以通络,又可协调机体之阴阳平衡,对脑源性疾病治疗有特效。本研究中,应用针刺联合计算机言语疾病矫治训练治疗观察组孤独症患儿,针刺选取头部相关穴位,其中舌三针可醒神通脑、利咽生津,改善患儿语言功能;本神、神庭穴与神志相关,为督脉气聚之所,针刺之可条畅神志、补益脑髓、清利头目;通里穴为少阴心经穴位,心开窍于舌,针刺本穴能疏通心经气血、

(C)1994-2024 China Academic Journal Electronic Publishing House. All rights reserved.　http://www.cnki.net

调神醒神,言语自出;百会穴能够调节脑腑经气,通达阴阳脉络,安神定志;言语一区、言语二区、言语三区属言语功能区。通过刺激以上诸穴,发挥补益心神、调神醒神之功,配合言语功能区,综合作用下改善孤独症患儿临床症状及异常行为、促进语言功能、认知功能恢复[20-21]。现代医学对针刺治疗的机制进行研究发现,一方面,无菌针直接刺激头部,能提高脑功能区周围神经组织兴奋性,激活位于语言中枢区域的神经元,解除大脑皮质抑制状态,加强语言相关器官与大脑皮层联系;另一方面,针刺促进脑侧支循环的建立,改善脑供血,促使部分休克的神经细胞功能恢复,改善孤独症患儿言语功能[22-23]。然而,本研究结果显示,对照组患儿治疗前后的CARS、GDS 评分比较差异均无统计学意义;提示单独使用计算机言语疾病矫治训练在孤独症患儿中的应用效果有限,原因可能与患儿理解能力欠佳、依从性差等因素有关,后续可进一步优化计算机言语疾病矫治训练方案。

4 结论

针刺联合计算机言语疾病矫治训练能够有效改善孤独症患儿临床症状及异常行为严重程度,促进语言功能及智力恢复,家属满意度较高,值得临床推广应用。

参考文献:

[1] GRANADILLO J L,P A STEGMANN A,GUO H,*et al*. Pathogenic variants in *TNRC6B* cause a genetic disorder characterised by developmental delay/intellectual disability and a spectrum of neurobehavioural phenotypes including autism and ADHD[J]. *J Med Genet*,2020,57(10):717-724.

[2] 徐秀,邹小兵,柯晓燕,等. 孤独症谱系障碍婴幼儿家庭实施早期干预专家共识[J]. 中国循证儿科杂志,2021,16(5):327-332.
XU X,ZOU X B,KE X Y,*et al*. Expert consensus on early intervention for infant families in autism spectrum disorder[J]. *Chin J Evid Based Pediatr*,2021,16(5):327-332.

[3] 朱梅,王雨晴. 孤独症谱系障碍及智力落后引起的语言发育迟缓儿童的共同注意力分析[J]. 蚌埠医学院学报,2021,46(12):1741-1743,1747.
ZHU M,WANG Y Q. Common attention analysis of children with language retardation caused by autism spectrum disorder and mental retardation[J]. *J Bengbu Med Coll*,2021,46(12):1741-1743,1747.

[4] 夏思颖,章丽雅,叶靓,等. 经颅直流电刺激联合计算机辅助综合训练对脑梗死后非流畅性失语症患者语言功能的改善效果[J]. 中国现代医生,2022,60(10):111-114.
XIA S Y,ZHANG L Y,YE J,*et al*. Effect of transcranial direct current stimulation combined with computerassisted comprehensive training on improving speech function of patients with non-fluency aphasia after cerebral infarction[J]. *China Mod Dr*,2022,60(10):111-114.

[5] 孙忠人,栾逸先,盛国滨,等. 针刺治疗痰湿中阻型轻度原发性高血压的临床疗效观察[J]. 中华中医药杂志,2020,35(1):451-454.
SUN Z R,LUAN Y X,SHENG G B,*et al*. Clinical observation on effects of acupuncture on mild essential hypertension in syndrome of phlegm-dampness blocking the middle[J]. *China J Tradit Chin Med Pharm*,2020,35(1):451-454.

[6] 崔我凛,张学丽. 麦粒灸结合针刺治疗急性期带状疱疹的临床疗效及其高频超声影像表现[J]. 世界中医药,2020,15(5):791-796.
CUI W L,ZHANG X L. Clinical effects of seed sized moxa cone combined with acupuncture on actue Herpes zoster and observation of its high-frequency ultrasound image[J]. *World Chin Med*,2020,15(5):791-796.

[7] 贾永男,顾建辉,魏清琳,等. 头穴透刺对孤独症谱系障碍儿童情绪和睡眠的影响[J]. 针刺研究,2021,46(11):948-952.
JIA Y N,GU J H,WEI Q L,*et al*. Effect of scalp acupuncture stimulation on mood and sleep in children with autism spectrum disorder[J]. *Acupunct Res*,2021,46(11):948-952.

[8] 中华医学会儿科学分会发育行为学组,中国医师协会儿科分会儿童保健专业委员会,儿童孤独症诊断与防治技术和标准研究项目专家组. 孤独症谱系障碍儿童早期识别筛查和早期干预专家共识[J]. 中华儿科杂志,2017,55(12):890-897.
DEVELOPMENTAL BEHAVIOR GROUP OF THE PEDIATRIC BRANCH OF THE CHINESE MEDICAL ASSOCIATION, THE CHILDREN'S HEALTH PROFESSIONAL COMMITTEE OF THE PEDIATRIC BRANCH OF THE CHINESE MEDICAL DOCTOR ASSOCIATION, THE EXPERT GROUP OF THE DIAGNOSIS AND PREVENTION TECHNOLOGY AND STANDARDS FOR CHILDREN'S AUTISM RESEARCH PROJECT. Expert consensus on early identification,screening,and intervention in children with autism spectrum disorders[J]. *Chin J Pediatr*,2017,55(12):890-897.

[9] RABEYRON T,ROBLEDO DEL CANTO J P,CARASCO E,*et al*. A randomized controlled trial of 25 sessions comparing music therapy and music listening for children with autism spectrum disorder[J]. *Psychiatry Res*,2020,293:113377.

[10] YAO D,ZENG Y,GAO M,*et al*. A research on developmental characteristics of children with language delay in Zhejiang Province,China[J]. *Front Pediatr*,2020,8:479.

[11] KAT S,XU L,GUO Y,*et al*. Reliability and validity of the simplified Chinese version of the aberrant behavior checklist in Chinese autism population[J]. *Front Psychiatry*,2020,11:545445.

[12] TAN S,ZHAO A,ZHANG J,*et al*. Toddler neurodevelopment is associated with ganglioside intake but not serum ganglioside[J]. *Asia Pac J Clin Nutr*,2020,29(3):584-592.

[13] PETERSON M J,WOERHLE T,HARRY M,*et al*. Family satisfaction in a neuro trauma ICU[J]. *Nurs Crit Care*,2022,27(3):334-340.

[14] 李恩耀,刘智慧,李立国,等. 孤独症谱系障碍儿童医教结合治疗模式的应用现状与思考[J]. 中国当代医药,2020,27(7):18-20.
LI E Y,LIU Z H,LI L G,*et al*. Application status and thinking of the medical and educational integration model for children with autism spectrum disorder[J]. *China Mod Med*,2020,27(7):18-20.

[15] 曹旭,李静,宋辉,等. 一个新的 TBR1 基因剪接变异导致的智力障碍伴自闭症和语言障碍家系分析及产前诊断[J]. 中华医学遗传学杂志,2021,38(10):933-936.
CAO X,LI J,SONG H,*et al*. Identification of a novel TBR1 gene

(C)1994-2024 China Academic Journal Electronic Publishing House. All rights reserved. http://www.cnki.net

variant in a Chinese pedigree affected with intellectual developmental disorder with autism and speech delay[J]. *Chin J Med Genet*,2021,38(10):933-936.

[16] 许晶莉,李程可,王嫱,等. 常规康复训练联合游戏治疗对孤独症谱系障碍共患注意缺陷多动障碍患儿的作用[J]. 临床精神医学杂志,2021,31(4):313-315.
XU J L,LI C K,WANG Q,*et al*. Effect of conventional rehabilitation combined with play therapy on autism spectrum disorders children co-morbid attention deficit hyperactivity disorder[J]. *J Clin Psychiatry*,2021,31(4):313-315.

[17] 吴佳,芮晓东,颜玲玲,等. 计算机言语疾病矫治训练在脑卒中并发失语症患者中的应用效果[J]. 反射疗法与康复医学,2021,2(20):115-117,121.
WU J,RUI X D,YAN L L,*et al*. Application effect of computer speech disease correction training in stroke patients with aphasia[J]. *RARM*,2021,2(20):115-117,121.

[18] 金炳旭,李诺,赵勇,等. 穴位埋线对自闭症儿童共同注意及社交沟通能力的影响:随机对照研究[J]. 中国针灸,2020,40(2):162-166.
JIN B X,LI N,ZHAO Y,*et al*. Effect of acupoint catgut embedding therapy on joint attention and social communication in children with autism spectrum disorder:a randomized controlled trial[J]. *Chin Acup Moxib*,2020,40(2):162-166.

[19] 刘盼,金瑄,赵宁侠,等. 刘晓萍对孤独症谱系障碍中医病因病机的认识[J]. 中华中医药杂志,2022,37(4):1999-2001.
LIU P,JIN X,ZHAO N X,*et al*. LIU Xiao-ping's understanding of traditional Chinese medicine etiology and pathogenesis of autism spectrum disorder[J]. *China J Tradit Chin Med Pharm*,2022,37(4):1999-2001.

[20] 黄哲真. 基于数据挖掘针刺治疗脑瘫腧穴经络规律研究[D]. 广州:广州中医药大学,2018.
HUANG Z Z. Research on law of meridian channels and collaters that are important for acupuncture treatment of cerebral palsy, based on data mining[D]. Guangzhou:Guangzhou University of Chinese Medicine,2018.

[21] 高峰,赵宁侠,张宁勃,等. 调神针法结合特殊教育和训练治疗孤独症谱系障碍儿童言语障碍临床疗效[J]. 中华中医药杂志,2019,34(12):5987-5989.
GAO F,ZHAO N X,ZHANG N B,*et al*. Clinical effect of Tiaoshen acupuncture combined with special education and training in treatment of speech disorder in children with autism spectrum disorder[J]. *China J Tradit Chin Med Pharm*,2019,34(12):5987-5989.

[22] 唐萍萍,许骞,陈栋,等. 头针刺激层次影响大脑皮层功能的机制探讨[J]. 针刺研究,2020,45(6):504-507.
TANG P P,XU Q,CHEN D,*et al*. Effect of scalp acupuncture stimulation on cerebral cortex function and related mechanism[J]. *Acupunct Res*,2020,45(6):504-507.

[23] 党伟利,李伟,马丙祥. 针刺改善孤独症谱系障碍患儿核心症状的临床观察[J]. 中国康复医学杂志,2020,35(5):527-532.
DANG W L,LI W,MA B X. Effects of acupuncture intervention on core symptoms in children with autism spectrum disorder[J]. *Chin J Rehabil Med*,2020,35(5):527-532.

(本文编辑:郭　潇)

(上接第 25 页)

[15] ROMERO-AROCA P, BAGET-BERNALDIZ M, PAREJA-RIOS A,*et al*. Diabetic macular edema pathophysiology:vasogenic versus inflammatory[J]. *J Diabetes Res*,2016,2016:2156273.

[16] MESQUIDA M,DRAWNEL F,FAUSER S. The role of inflammation in diabetic eye disease[J]. *Semin Immunopathol*,2019,41(4):427-445.

[17] 张劲,明媚. 增生性糖尿病视网膜病变患者玻璃体、房水和血浆中 VEGF 表达与 IL-6,IL-8 和 TNF-α 水平的相关性研究[J]. 现代检验医学杂志,2021,36(4):55-59.
ZHANG J,MING M. Correlation of VEGF expression with IL-6, IL-8 and TNF-α levels in vitreous body,aqueous humor and plasma of patients with proliferative diabetic retinopathy[J]. *J Mod Lab Med*,2021,36(4):55-59.

[18] FUNATSU H,NOMA H,MIMURA T,*et al*. Association of vitreous inflammatory factors with diabetic macular edema[J]. *Ophthalmology*,2009,116(1):73-79.

[19] ZHU Y,CAI Q,LI P,*et al*. The relationship between peripheral blood inflammatory markers and diabetic macular edema in patients with severe diabetic retinopathy[J]. *Ann Palliat Med*,2022,11(3):984-992.

[20] DASCALU A M,SERBAN D,TANASESCU D,*et al*. The value of white cell inflammatory biomarkers as potential predictors for diabetic retinopathy in type 2 diabetes mellitus (T2DM)[J]. *Biomedicines*,2023,11(8):2106.

[21] BING R S,TSUI W L,DING D C. The association between diabetes mellitus, high monocyte/lymphocyte ratio, and survival in endometrial cancer:a retrospective cohort study[J]. *Diagnostics*,2022,13(1):44.

[22] WANG H, GUO Z, XU Y. Association of monocyte-lymphocyte ratio and proliferative diabetic retinopathy in the U. S. population with type 2 diabetes[J]. *J Transl Med*,2022,20(1):219.

[23] CUI T,WANG C,ZHU Q,*et al*. Association between neutrophil percentage-to-albumin ratio and 3-month functional outcome in acute ischemic stroke patients with reperfusion therapy[J]. *Front Neurol*,2022,13:898226.

[24] ZHANG H,WU T,TIAN X,*et al*. High neutrophil percentage-to-albumin ratio can predict occurrence of stroke-associated infection[J]. *Front Neurol*,2021,12:705790.

[25] VURAL E,HAZAR L. Assessment of inflammation biomarkers in diabetic macular edema treated with intravitreal dexamethasone implant[J]. *J Ocul Pharmacol Ther*,2021,37(7):430-437.

(本文编辑:李胜利)

(C)1994-2024 China Academic Journal Electronic Publishing House. All rights reserved.　http://www.cnki.net